ALFIDO TECH INTERNSHIP

Deep Learning Image Classification using Transfer Learning

Prepared for Alfido Tech Artificial Intelligence Internship

Abstract

This project implements an image classification system using Deep Learning and Transfer Learning techniques using the CIFAR-10 dataset and a pretrained ResNet18 model.

Objective

Develop an image classifier using transfer learning, data augmentation, model evaluation, and model persistence.

Dataset Description

CIFAR-10 dataset containing 60,000 images across 10 classes. 50,000 images are used for training and 10,000 for testing.

Tools and Technologies

Python, PyTorch, Torchvision, NumPy, Matplotlib, Scikit-Learn, Jupyter Notebook/VS Code.

Methodology

Data preprocessing, augmentation, transfer learning with ResNet18, model training, evaluation, and saving the trained model.

Model Architecture

Pretrained ResNet18 with the final fully connected layer replaced for 10-class classification.

Training Results

Expected accuracy range: 75–85% depending on training conditions and hardware.

Evaluation Metrics

Accuracy, Precision, Recall, and F1-Score are used for performance evaluation.

Artifacts Generated

cifar10_resnet18_model.pth, training_curves.png, image_classifier.py, README.md, classification report screenshots.

Responsible AI Considerations

No personal data is used. The project emphasizes transparency, documentation, and educational usage.

Future Enhancements

Use larger models, train longer, deploy with Flask/FastAPI, and optimize hyperparameters.

Conclusion

The project demonstrates practical deep learning skills using transfer learning and satisfies the internship requirements.